Beyond One Granularity: Spectral Ensemble and Consensus Expansion for Training-Free Composed Image Retrieval

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Abstract

Composed image retrieval (CIR) enables expressive multimodal search: given a reference image and a text modifier, the goal is to retrieve images that match the visual content as altered by the text. Building effective CIR systems has historically required expensive triplet-annotated training data, limiting scalability. Training-free CIR sidesteps this bottleneck by repurposing pre-trained vision-language models to compose queries without any task-specific supervision. However, existing training-free methods commit to a single, globally fixed projection granularity when aligning visual and textual features. This creates an inherent trade-off: coarse granularities capture broad categorical distinctions but lose fine-grained detail, while fine granularities recover instance-level attributes but amplify noise. The problem is especially acute in instance-level CIR, where per-query databases vary from fewer than 10 to over 10,000 images, making any fixed setting suboptimal across the evaluation. We therefore propose a unified retrieval method with two complementary components that fuse scores across multiple granularities to capture complementary evidence and yield more robust rankings. MACE (Multi-granularity Adaptive Composition with Expansion) constructs a spectral ensemble from multiple nested truncation levels and fuses their scores without any additional eigendecomposition or database modification. CGCQE (Cross-Granularity Consensus Query Expansion) integrates into MACE to strengthen pseudo-relevance feedback, selecting expansion candidates that rank consistently well across all granularities and preventing granularity-specific false positives from corrupting the expanded query. Experiments on five CIR benchmarks, including the challenging instance-level i-CIR, show that MACE with CGCQE achieves state-of-the-art performance, outperforming both training-free and supervised zero-shot methods.

CCS Concepts

• **Information systems → Information retrieval**; **Language models**; • **Computing methodologies → Artificial intelligence**.

Keywords

Composed Image Retrieval; Multimodal Learning; Information Retrieval

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1 Introduction

Composed image retrieval integrates visual and textual queries to retrieve target images. Given a reference image and a text specifying desired modifications, a retrieval system locates database images that satisfy both the visual identity and the textual intent. This paradigm applies broadly, from product search to creative exploration. Pre-trained vision-language models [23, 27] have enabled training-free methods that operate directly on frozen features without task-specific supervision, making them scalable and easy to deploy in new domains.

A key challenge is the modality gap: image and text representations occupy distinct regions of the shared embedding space even after contrastive pre-training [7]. Contrastive principal component analysis [1, 4, 25] addresses this by projecting image features onto

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the leading eigenvectors of a contrastive covariance matrix, amplifying object-related variance while suppressing style-related variation. The projected features align more closely with text features, enabling reliable cross-modal similarity computation.

Despite its utility, fixing a single spectral truncation level introduces an unavoidable trade-off. Retaining fewer components preserves only dominant discriminative directions, ensuring robust coarse-level matching at the cost of fine visual detail [32]. Retaining more components recovers subtle variations but also includes noisy or less transferable spectral elements. Committing to one truncation level globally is inadequate for the diverse retrieval conditions encountered in practice, especially in instance-level settings where individual databases range from one to over ten thousand images [25].

To overcome this bottleneck, we propose Multi-granularity Adaptive Composition with Expansion (MACE). The method evaluates retrieval candidates at multiple spectral truncation levels and fuses the resulting scores. Rankings produced at different granularities exhibit low mutual correlation, meaning each spectral slice captures visual information the others miss. Because all projections are nested column subsets of a single eigen-decomposition, the ensemble operates entirely on the query side with no additional decomposition or database modifications.

We also address the reliability of pseudo-relevance feedback in this multi-granularity context through Cross-Granularity Consensus Query Expansion (CGCQE). Standard query expansion independently selects top-ranked candidates at each granularity, risking the integration of granularity-specific false positives. Our method instead aggregates similarities across all granularities and selects only candidates that rank consistently well under diverse spectral projections. This consensus pool effectively prevents unreliable items from corrupting the expanded query.

The main contributions of our work are summarized as follows:

(1) We identify the structural limitation of single-granularity projection in training-free composed image retrieval and propose MACE, a multi-granularity score fusion framework designed to capture hierarchical visual similarities.
(2) We propose CGCQE, a robust candidate selection mechanism that filters unreliable expansion items by enforcing cross-granularity consensus, improving pseudo-relevance feedback across heterogeneous databases.
(3) We conduct extensive evaluations across five benchmarks and provide systematic ablations that reveal practical insights for expansion design. Specifically, we show that per-granularity independent candidate selection introduces unreliable expansion items, and that text-guided expansion causes cross-modal calibration mismatches.

2 Related Work

2.1 Composed Image Retrieval

Composed image retrieval (CIR) requires a system to retrieve target images by jointly processing a reference image and a textual modification query [37]. Early supervised approaches encode this query pair through specialized fusion modules trained end-to-end on labeled triplets. TIRG [34] introduces residual gating to compose visual and textual features, while subsequent work such as CLIP4CIR [4] adapts contrastive vision-language models to the task through lightweight combiner networks trained on CIR-specific annotations. Several benchmarks have been established to evaluate these methods, including FashionIQ [35], CIRR [22], and CIRCO [3]. The recently introduced i-CIR setting extends evaluation to the instance level, where each query is evaluated against a dedicated database of variable size [25], substantially increasing the difficulty for methods that commit to a single retrieval strategy.

To reduce dependence on labeled triplets, training-free approaches leverage the zero-shot generalization of pre-trained vision-language models. Pic2Word [29] and SEARLE [3] convert the reference image into pseudo-word tokens via textual inversion, enabling composition entirely in text space without task-specific supervision. LinCIR [12] simplifies this further with a linear projection that maps image embeddings to textual directions. CIReVL [17] introduces contrastive principal component analysis to project visual features onto semantically meaningful subspaces, establishing a strong training-free baseline. Our work extends this spectral projection paradigm by relaxing its single-granularity constraint, demonstrating that ensembling complementary spectral truncation levels yields consistent gains across all benchmarks.

Large multimodal models have spurred a new generation of training-free CIR methods that leverage language model reasoning to bridge the semantic gap between reference images and textual modifiers. Context-I2W [31] generates context-sensitive pseudo-word tokens by conditioning textual inversion jointly on the visual content of the reference and the modifier text, addressing the context-agnostic limitation of earlier inversion methods. Instruction-tuned models such as LLaVA [21] and BLIP-2 [18] have been adapted to reformulate CIR as a captioning problem: given the reference image and modifier, the model generates a natural-language description of the intended target for zero-shot text-to-image retrieval. MagicLens [39] extends this approach by training a unified retrieval backbone through self-supervised instruction tuning over diverse natural-language retrieval intents.

2.2 Vision-Language Models and Feature Alignment

Large-scale vision-language models pre-trained with contrastive objectives [36] have become the foundation of modern retrieval systems. CLIP [27] and ALIGN [16] learn joint image-text embeddings by aligning paired samples while contrasting unpaired ones, producing feature spaces where cross-modal similarity can be measured by dot product. A persistent challenge is the modality gap: even after contrastive pre-training, image and text embeddings systematically occupy distinct regions of the shared space [20]. Contrastive principal component analysis [1] addresses this by projecting image features onto the leading eigenvectors of a difference covariance matrix, computed between an object corpus and a style corpus. This suppresses stylistic variance while amplifying object-relevant dimensions, yielding image projections that align more closely with text features.

Subsequent advances in vision-language pre-training have introduced stronger backbones for retrieval. SigLIP [38] replaces the softmax in CLIP's contrastive loss with a sigmoid formulation that enables per-pair positive-negative interactions, yielding tighter cross-modal alignment without requiring large batch sizes.

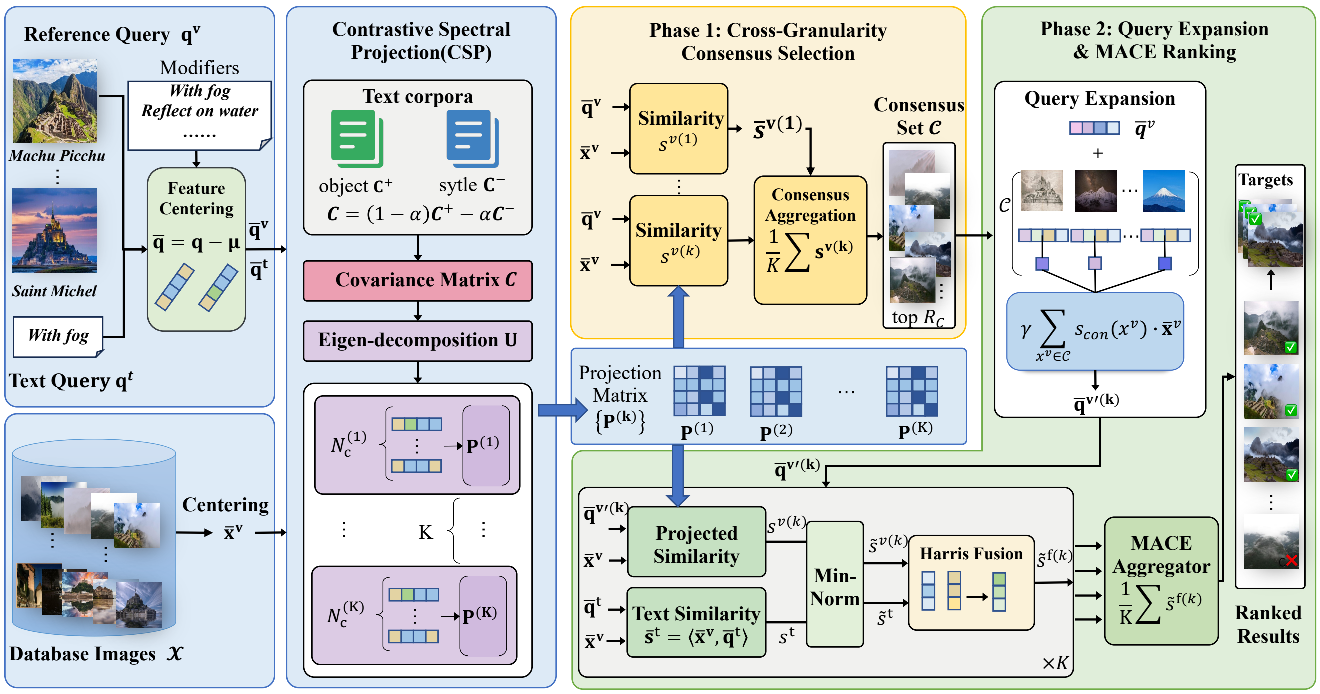


Figure 1: Overview of the MACE architecture. Cross-granularity consensus filters expansion candidates by spectral consistency. Per-granularity expansion then feeds into spectral hierarchy aggregation for the final ranking.

DINOv2 [23] shows that purely self-supervised training on curated visual data produces general-purpose features competitive with those from language-supervised models, revealing that rich spectral structure exists even in representations learned without language grounding.

2.3 Query Expansion in Image Retrieval

Pseudo-relevance feedback enriches a query representation using top-ranked retrieval results, under the assumption that high-scoring candidates are likely relevant. Average Query Expansion (AQE) [5] averages the descriptors of the top-k results and uses the aggregate as a refined query. αQE [26] extends this by weighting candidate contributions by their similarity scores raised to a power α, reducing the influence of marginal matches. Database-side Augmentation (DBA) [2] applies the same principle symmetrically to database entries rather than queries, and attention-based variants further improve robustness by weighting candidate features according to learned relevance [10].

Despite their effectiveness in single-modal retrieval, these methods are fragile when top-ranked candidates include false positives. In the CIR setting this fragility is amplified by database size heterogeneity and cross-modal calibration mismatches. More recent work on query expansion has moved beyond fixed descriptor averaging toward learned relevance weighting and multi-system score fusion. Reciprocal rank fusion [6] merges ranked lists from multiple retrieval systems by aggregating reciprocal rank positions, offering robustness to score scale differences across systems. It nonetheless discards relative confidence magnitudes between candidates and performs sub-optimally when retrieval scores carry meaningful similarity information, as in our score-level ensemble.

3 Method

3.1 Preliminary

Task formulation. In composed image retrieval (CIR), a query consists of a reference image q^v and a text modifier q^t describing a desired attribute change. Given a database $\mathcal{X} = \{x_1^v, \ldots, x_n^v\}$, the objective is to rank images such that the target $x^* \in \mathcal{X}$, whose visual content corresponds to q^v modified as described by q^t, is placed highest. Formally, a scoring function $s(q^v, q^t, x^v) \in \mathbb{R}$ must satisfy $s(q^v, q^t, x^*) > s(q^v, q^t, x_j^v)$ for all $x_j^v \neq x^*$.

Training-free setting. We operate without task-specific supervision: no annotated triplets are used and no model parameters are fine-tuned. A frozen vision-language model (VLM), specifically CLIP [27], provides encoders $\phi^v : \mathcal{I} \to \mathbb{R}^d$ and $\phi^t : \mathcal{T} \to \mathbb{R}^d$ that map images and text into a nominally shared d-dimensional space. Despite this shared space, the two modalities occupy geometrically separated regions, a phenomenon known as the *modal gap* [20] that makes naive dot-product composition unreliable. Our method operates entirely on frozen embeddings and is therefore independent of annotation availability and domain-specific fine-tuning.

Instance-level evaluation. We target the instance-level evaluation protocol [25], where each query (q_i^v, q_i^t) is paired with a dedicated per-query database $\mathcal{X}_i$ of varying size n_i. Unlike category-level benchmarks that share a global gallery, this setting reflects realistic retrieval scenarios where the effective search space is unpredictable. The pronounced variability in n_i makes a globally fixed projection dimensionality suboptimal, and directly motivates our multi-granularity framework.

Notation. All image and text features are centered by subtracting pre-computed modality means:

$$\bar{\mathbf{q}}^v = \phi^v(q^v) - \boldsymbol{\mu}^v, \quad \bar{\mathbf{q}}^t = \phi^t(q^t) - \boldsymbol{\mu}^t, \quad \bar{\mathbf{x}}^v = \phi^v(x^v) - \boldsymbol{\mu}^v. \tag{1}$$

Centered features are denoted with a bar throughout. To achieve better scalability and generalization ability, we draw on the method proposed in [25] and estimate $\boldsymbol{\mu}^v$ using a large external dataset $\mathcal{X}^v$, such as LAION [30]. In the same way, we compute the text feature mean $\boldsymbol{\mu}^t$ on a predefined textual corpus (introduced in Section 3.2) that covers semantically related concepts and content. This centering removes modality-specific distributional biases and is a prerequisite for the contrastive spectral projection described next.

3.2 Contrastive Spectral Projection and Modality Composition

Training-free CIR models typically evaluate image-to-image and text-to-image similarities independently before composing them. Building on the centered features $\bar{\mathbf{q}}^v, \bar{\mathbf{q}}^t, \bar{\mathbf{x}}^v$ defined in Equation 1, we adopt a pipeline comprising contrastive spectral projection, min-normalization, and cross-modal composition via the Harris criterion.

Subsequently, a contrastive covariance matrix $\mathbf{C} = (1-\alpha)\mathbf{C}^+ - \alpha\mathbf{C}^-$ is constructed using text features from an object corpus ($\mathbf{C}^+$) and a style corpus ($\mathbf{C}^-$). Eigendecomposition of $\mathbf{C}$ yields an orthogonal matrix $\mathbf{U} \in \mathbb{R}^{d\times d}$ whose columns are eigenvectors sorted by descending eigenvalue. Selecting the top N_c constructs a projection matrix $\mathbf{P} \in \mathbb{R}^{d\times N_c}$ that amplifies object-related dimensions while suppressing style.

Similarity is evaluated in this projected subspace for image features and the original centered space for text:

$$s^v = \langle \mathbf{P}^\top \bar{\mathbf{x}}^v,\ \mathbf{P}^\top \bar{\mathbf{q}}^v \rangle, \quad s^t = \langle \bar{\mathbf{x}}^v,\ \bar{\mathbf{q}}^t \rangle. \tag{2}$$

The similarities are then affinely rescaled to map the empirical minimum to zero. The normalized scores $\tilde{s}^v$ and $\tilde{s}^t$ are fused via the Harris criterion [13]:

$$\tilde{s}^f = \tilde{s}^v \tilde{s}^t - \lambda(\tilde{s}^v + \tilde{s}^t)^2, \tag{3}$$

where λ acts as a balancing parameter. The product term rewards joint relevance, whereas the penalty term mitigates single-modality false positives. A fixed N_c constrains retrieval to a single operating point, a limitation our method directly addresses.

3.3 Multi-Granularity Spectral Ensemble

We define granularity as the number of retained eigencomponents N_c in contrastive principal component analysis. A small N_c corresponds to coarse-grained matching (category-level), while a large N_c corresponds to fine-grained matching (instance-level). The spectrum derived from contrastive PCA [1] intrinsically encodes a hierarchy of visual information. The leading eigenvectors, corresponding to a small N_c, capture dominant object-versus-style contrasts critical for broad category separation. Conversely, trailing eigenvectors encode progressively finer contrasts that benefit instance-level discrimination. However, no single truncation point N_c is universally optimal: a small value may discard fine-grained signal, while a large value may include noisy dimensions that generalize poorly across diverse query conditions.

Algorithm 1 MACE with CGCQE

Require: Query image q^v, query text q^t, database $\mathcal{X} = \{x_i^v\}_{i=1}^n$; granularities $\mathcal{N} = \{N_c^{(1)}, \ldots, N_c^{(K)}\}$; hyperparameters $\alpha, \lambda, \gamma, R_c$

Ensure: Database $\mathcal{X}$ ranked by s_{MACE}

```
Center all features: q̄^v, q̄^t, {x̄_i^v}   ▷ subtract modality means
Compute C = (1−α)C⁺ − αC⁻; eigendecompose C → U
Compute s_i^t = ⟨x̄_i^v, q̄^t⟩ for all i; min-normalize → s̃_i^t
// Phase 1 — Cross-Granularity Consensus
for k = 1 to K do
    P^(k) ← first N_c^(k) columns of U
    s_i^v(k) ← ⟨P^(k)⊤ x̄_i^v, P^(k)⊤ q̄^v⟩ for all i
end for
s_con,i ← (1/K) Σ_{k=1}^K s_i^v(k)   ▷ Eq. 5
C ← top-R_c({s_con,i})
// Phase 2 — Per-Granularity Expansion and Ensemble
for k = 1 to K do
    q̄^v′(k) ← q̄^v + γ Σ_{x^v∈C} s_con(x^v) · x̄^v; ℓ2-normalize   ▷ Eq. 6
    s_i^v(k) ← ⟨P^(k)⊤ x̄_i^v, P^(k)⊤ q̄^v′(k)⟩ for all i; min-normalize → s̃_i^v(k)
    s̃_i^f(k) ← s̃_i^v(k) s̃_i^t − λ(s̃_i^v(k) + s̃_i^t)^2   ▷ Eq. 3
end for
s_MACE,i ← (1/K) Σ_{k=1}^K s̃_i^f(k)   ▷ Eq. 4
return X ranked by {s_MACE,i}
```

Our spectral ensemble fuses retrieval scores from multiple nested PCA projections with different N_c. Here, the spectral hierarchy refers to the ordered eigenvectors arranged from coarse to fine visual patterns. Rather than selecting a single N_c, which inevitably sacrifices either coarse-grained or fine-grained discriminability, MACE systematically traverses the spectral hierarchy by evaluating a set of granularities $\mathcal{N} = \{N_c^{(1)}, \ldots, N_c^{(K)}\}$. For each $N_c^{(k)}$, the projection $\mathbf{P}^{(k)} \in \mathbb{R}^{d\times N_c^{(k)}}$ comprises the first $N_c^{(k)}$ columns of a single shared eigendecomposition. Each granularity produces a composite similarity $\tilde{s}^{f(k)}$ via Equation 3, and the final ensemble score is obtained by averaging across all granularities:

$$s_{\text{MACE}} = \frac{1}{K}\sum_{k=1}^{K} \tilde{s}^{f(k)}. \tag{4}$$

Aggregating over the spectral hierarchy exploits complementary information: coarser projections reliably separate object categories, while finer ones resolve instance-level attributes. Because all projections are nested column subsets of a single shared eigendecomposition and operate on the same embedding space, their similarity scores are natively commensurate and require no inter-system calibration. A candidate that scores consistently well across multiple granularities is more likely to be a true positive, while one that peaks under a single $N_c^{(k)}$ due to accidental eigenvector alignment is diluted in the averaged result. Spectral diversity thus acts as an implicit regularizer, improving robustness without any additional decomposition or modification of database embeddings.

3.4 Cross-Granularity Consensus Query Expansion

To enrich the query for each granularity, we incorporate pseudo-relevance feedback using image-only expansion weights. Text similarity is excluded from the expansion step to avoid cross-modal calibration mismatches, as image-to-image and text-to-image dot products lie in incompatible numerical ranges. Text contributes only at the final Harris fusion stage. We propose **Cross-Granularity Consensus Query Expansion (CGCQE)**, whose core idea is to select only candidates that perform well across all granularities (called the consensus set), while rejecting candidates that are high in a single granularity but low in others. This consensus filtering eliminates unreliable items and stabilizes the expanded query.

Let R denote the number of expansion candidates selected per granularity in standard query expansion, and R_c ($R_c < R$) denote the size of the consensus candidate set shared across all granularities. A naive approach selects the top-R candidates independently at each granularity k, using per-projection similarity as the expansion weight. This is fragile: a candidate may score highly under a specific $\mathbf{P}^{(k)}$ due to accidental eigenvector alignment, corrupting the expanded query for that granularity. CGCQE resolves this by extending the spectral complementarity principle to candidate selection. Applying the same cross-granularity averaging as MACE, we compute a consensus score for each database image:

$$s_{\text{con}}(x^v) = \frac{1}{K}\sum_{k=1}^{K}\langle \mathbf{P}^{(k)\top}\bar{\mathbf{x}}^v,\ \mathbf{P}^{(k)\top}\bar{\mathbf{q}}^v\rangle. \tag{5}$$

The top-R_c items by s_{con} form the consensus set C, shared uniformly across all granularities for expansion:

$$\bar{\mathbf{q}}^{v\prime(k)} = \bar{\mathbf{q}}^v + \gamma \sum_{x^v\in C} s_{\text{con}}(x^v)\cdot \bar{\mathbf{x}}^v, \tag{6}$$

after which $\bar{\mathbf{q}}^{v\prime(k)}$ is ℓ_2-normalized. Only candidates consistently retrieved across the full spectral hierarchy enter C ($R_c < R$), ensuring the expansion vector captures genuinely shared visual evidence rather than artifacts of any single projection.

4 Experiments

4.1 Experimental Setup

Datasets and evaluation protocol. We evaluate on five CIR benchmarks spanning two paradigms. For *instance-level* retrieval we use **i-CIR** [25], which contains 202 object instances, 1,883 composed queries, and 750K images. Each instance has its own dedicated per-query database (median 3,420 hard negatives per instance), making it substantially more challenging than benchmarks with a shared global gallery. Following [25] we report *macro-mAP*: mean Average Precision (AP) is first averaged over all queries of each instance to obtain a per-instance mAP, and these per-instance values are then averaged across all instances. For *domain-conversion* retrieval we use four benchmarks: **ImageNet-R** [14] (30K images, 200 classes, five style domains), **MiniDomainNet** (MiniDN, [41]) (140K images, 126 classes, four domains), **NICO++** [40] (89K images, 60 categories, six contexts), and **LTLL** [9] (500 images, 25 landmark locations, historic/modern split). On all domain-conversion datasets retrieval performance is measured by mAP.

Baselines. We include four unimodal or score-level fusion baselines: **Text** scores each database image by the text-to-image dot product; **Image** by the image-to-image dot product; **Text + Image** by their sum; and **Text × Image** by their product. For supervised zero-shot CIR methods we compare against **Pic2Word** [29], **CompoDiff** [11], **CIReVL** [17], **SEARLE** [3], **MCL** [19], **MagicLens** [39], and **CoVR-2** [33]. For training-free methods we compare against **WeiCom** [24], **FreeDom** [8], and **BASIC** [25]. All methods use CLIP [27] ViT-L/14 as the backbone. CompoDiff alone uses the larger ViT-G/14. We denote the variant *without* query expansion with †.

Implementation details. We use CLIP ViT-L/14 as the frozen backbone throughout. The contrastive covariance matrix is built from the same object corpus C^+ (1,800 entries) and style corpus C^- (1,000 entries) used by BASIC [25], both generated via ChatGPT [15] with identical prompts. Modality means $\boldsymbol{\mu}^v$ and $\boldsymbol{\mu}^t$ are estimated on LAION [30]. Min-normalization statistics ($s^v_{\min}$, $s^t_{\min}$) are derived from a synthetic dataset generated by Stable Diffusion [28], following [25]. The granularity set is $\mathcal{N} = \{N_c^{(1)}, \ldots, N_c^{(K)}\}$ with $K = 5$ levels spaced as $\{50, 100, 150, 200, 250\}$. The consensus candidate set size is $R_c = 15$. The expansion weight is $\gamma = 0.4$. The contrastive mixing coefficient $\alpha = 0.2$ and Harris regularization weight $\lambda = 0.1$ are carried over from BASIC without re-tuning. All hyperparameters were fixed once on the i-CIR$_{\text{dev}}$ split (15 instances, 92 queries, 45K images) provided by [25] and are held constant across all datasets and experiments. No database embeddings are modified. All computation is query-side only.

4.2 Comparison with State of the Art

Table 1 compares MACE with all competitors on five benchmarks. MACE achieves state-of-the-art performance on every dataset, surpassing all training-free and supervised zero-shot CIR methods without any task-specific training or database re-encoding.

Gains over the training-free baseline. Against BASIC (the strongest prior training-free method), MACE improves by +3.01, +5.81, +3.81, +3.04, and +6.92 mAP on ImageNet-R, NICO++, MiniDN, LTLL, and i-CIR, respectively, for an average gain of +4.52. The largest absolute improvement is on i-CIR, which rises from 34.35 to 41.27 mAP. As the only instance-level benchmark in our evaluation, i-CIR demands precisely the spectral diversity that MACE provides: resolving fine-grained intra-class variation across diverse object instances requires the complementary cues that individual projections cannot simultaneously capture.

Gains over supervised methods. Despite requiring no training, MACE outperforms all supervised zero-shot CIR competitors by substantial margins. On i-CIR, MACE's score of 41.27 exceeds the strongest trained competitor, CoVR-2, by 12.77 mAP. On ImageNet-R, MACE leads the best trained result, CIReVL, by 12.44 mAP.

Table 1: Average mAP (%) comparison across five CIR benchmarks. All methods use CLIP ViT-L/14; CompoDiff uses ViT-G/14 (∗). Bold: best overall. *Note*: i-CIR reports macro-mAP (per-instance mAP averaged across instances).

Method	Training-free	ImageNet-R	NICO++	MiniDN	LTLL	i-CIR	Avg.
Text	✓	0.74	1.09	0.57	5.72	3.01	2.23
Image	✓	3.84	6.32	6.66	16.49	3.04	7.27
Text + Image	✓	6.21	9.30	9.33	17.86	8.20	10.18
Text × Image	✓	7.83	9.79	9.86	23.16	17.48	13.62
WeiCom [24]	✓	10.47	10.54	8.52	26.60	18.03	14.83
Pic2Word [29]	–	7.88	9.76	12.00	21.27	19.36	14.05
CompoDiff* [11]	–	12.88	10.32	22.95	21.61	9.63	15.48
CIReVL [17]	–	18.11	17.80	26.20	32.60	18.66	22.67
SEARLE [3]	–	14.04	15.13	21.78	25.46	19.90	19.26
MCL [19]	–	8.13	19.09	18.41	16.67	19.89	16.44
MagicLens [39]	–	9.13	19.66	20.06	24.21	27.35	20.08
CoVR-2 [33]	–	11.52	24.93	27.76	24.68	28.50	23.48
FreeDom [8]	✓	25.81	23.24	32.14	30.82	15.76	25.55
BASIC [25]	✓	27.54	28.90	35.75	38.22	34.35	32.95
MACE (ours)	✓	**30.55**	**34.71**	**39.56**	**41.26**	**41.27**	**37.47**

MACE leads on every dataset, confirming that training-free multi-granularity spectral aggregation generalises more broadly than models optimised on fixed triplet distributions.

4.3 Component Contributions

Table 2 isolates the contribution of each proposed component, starting from the BASIC† single-granularity baseline [25].

Multi-granularity ensemble (MACE†). Replacing the fixed single projection at N_c=250 with the five-level ensemble, while keeping all other components identical and omitting any query expansion, yields consistent gains across all datasets, averaging +1.80 mAP. The effect is most pronounced on i-CIR, where performance climbs from 34.35 to 39.98, a gain of 5.63 mAP: instance-level retrieval benefits strongly from fusing complementary semantic granularities, as coarser projections capture broad object identity while finer ones resolve discriminative local attributes. Gains on domain-conversion benchmarks are smaller but reliable, at 1.08 mAP on ImageNet-R and 1.57 on MiniDN, suggesting that spectral diversity also improves style-invariant matching even when cross-domain generalisation is the primary challenge.

Independent per-granularity query expansion (+Indep. QE). Adding query expansion where each granularity independently selects its own top-R_c candidates further improves most datasets by an average of +0.86 mAP, but causes a slight regression on ImageNet-R, which drops from 28.62 to 28.29. This reveals a key failure mode of naive multi-granularity expansion: coarser projections may retrieve stylistically similar but semantically inconsistent candidates, which contaminate the expansion pool and degrade style-invariant matching.

Consensus-guided expansion (CGCQE). Replacing per-granularity candidate selection with the shared consensus set C, which retains only images ranking highly across *all* granularities, delivers the largest and most consistent improvements, averaging +1.86 mAP over the independent QE baseline. Most strikingly, ImageNet-R recovers and surpasses the no-expansion baseline, rising from 28.29 to 30.55 mAP, directly confirming that consensus filtering eliminates the outlier candidates introduced by individual granularities. NICO++ and LTLL show particularly large gains, at 2.94 and 2.12 mAP respectively, indicating that consensus selection is especially critical under large domain shifts, where isolated granularity perspectives are most susceptible to domain-specific noise.

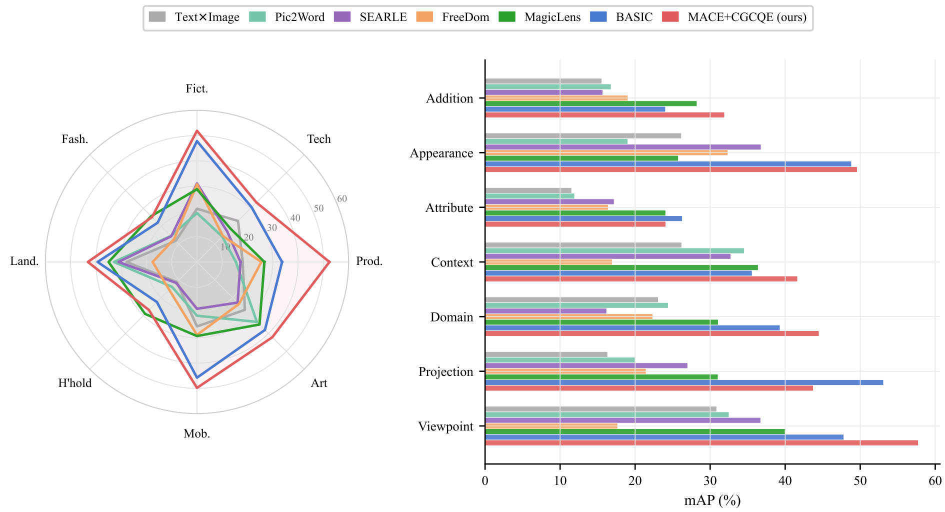
Table 2: Component ablation: mAP (%) on all five benchmarks. BASIC: single-granularity baseline [25]. MACE: multi-granularity ensemble, no expansion. +Indep. QE: per-granularity independent candidate selection. +CGCQE: consensus candidate set (full method).

Method	QE	IN-R	NICO	Mini	LTLL	i-CIR	Avg.
BASIC† [25]	×	27.54	28.90	35.75	38.22	34.35	32.95
MACE† (ours)	×	28.62	29.01	37.32	38.83	39.98	34.75
+Indep. QE	✓	28.29	31.77	37.94	39.14	40.91	35.61
+CGCQE (ours)	✓	**30.55**	**34.71**	**39.56**	**41.26**	**41.27**	**37.47**

4.4 Granularity Set Design

We study how the choice of granularity set $\mathcal{N}$ affects performance. Figure 3 presents two analyses: (a) the number of granularities K (levels equally spaced from 50 to 250), and (b) the maximum projection dimension $N_c^{(K)}$ (fixing K=5 while sweeping from 100 to 400).

Effect of K. Performance increases consistently with K across all five datasets and saturates at K=5. The gain from K=1 to K=5 is most pronounced on i-CIR, the only instance-level benchmark, where aggregating diverse spectral views provides the fine-grained discriminative cues required to distinguish individual object instances from a large pool of hard negatives. Adding further granularities beyond K=5 yields negligible improvement, confirming



(a) Visual category

(b) Textual modification category

Figure 2: Per-category mAP (%) on i-CIR. (a) Primary visual categories (8 groups). (b) Textual modification categories (7 groups). Competitor results from [25].

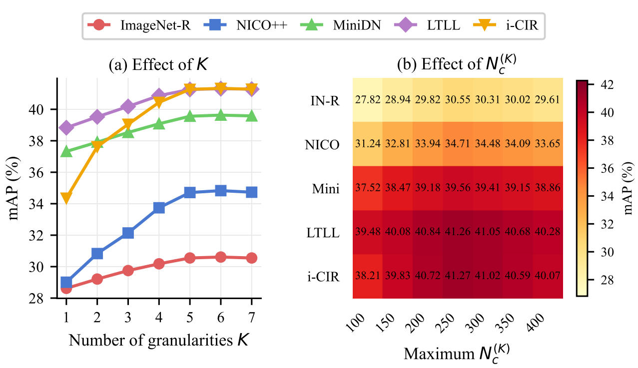


Figure 3: Sensitivity to granularity set design. (a) mAP vs. number of granularities K (equal spacing, $N_c^{(K)}$=250). (b) Heatmap of mAP across datasets and maximum projection dimension $N_c^{(K)} \in \{100, \ldots, 400\}$ (K=5).

that five equally-spaced levels spanning $[50, 250]$ suffice to capture the complementary benefit of multi-granularity fusion.

Effect of $N_c^{(K)}$. The heatmap shows that N_c=250 is globally near-optimal, with performance degrading on both sides. At low N_c (100–150), projections discard too many informative eigenvectors. The penalty is largest on i-CIR and LTLL, where fine-grained and cross-temporal matching require rich spectral resolution. At high N_c (300–400), the projection begins to retain noisy or style-correlated components that weaken the contrastive effect, causing a mild but consistent drop on domain-conversion benchmarks. The safe operating range $N_c \in [200, 300]$ encompasses our default $N_c^{(K)}$=250, motivating this choice as a robust mid-point.

4.5 Sensitivity Analysis

Figure 5 jointly examines the effect of the consensus threshold R_c and the one-factor-at-a-time sensitivity of the three hyperparameters α, λ, and γ.

Consensus threshold R_c. Panel (a) sweeps $R_c \in \{5, 10, 15, 20, 30, 50\}$, comparing CGCQE (solid lines) with per-granularity independent QE (dashed lines). CGCQE outperforms independent QE across the entire range, demonstrating that the advantage of consensus-guided candidate selection is structural rather than threshold-specific. The gap is largest at small R_c (5–10), where independent QE suffers from high variance in granularity-specific top lists, while the consensus filter remains effective even with a tight budget. Performance peaks near R_c=15 and remains stable in $[10, 20]$. Beyond this range, admitting more candidates allows per-granularity outliers to re-enter the expansion pool and mildly degrades accuracy. We adopt R_c=15 as the default.

Contrastive mixing weight α. Panel (b) sweeps $\alpha \in [0.0, 0.8]$ while fixing λ=0.1 and γ=0.4. Performance is low at α=0.0, reflecting the importance of attenuating style-correlated eigenvectors, and rises steeply as α increases. All datasets plateau broadly in $\alpha \in [0.15, 0.35]$ and decay gradually beyond 0.4, where over-suppression begins to remove semantically informative components. Our default α=0.2 sits near the common optimum across all benchmarks.

Harris regularization weight λ. Panel (c) sweeps $\lambda \in [0.000, 0.200]$. The curves rise quickly from λ=0 and flatten for $\lambda \in [0.05, 0.15]$, with the global peak consistently near λ=0.10. The stability of this plateau across datasets confirms that Harris regularization acts as a universal stabilizer balancing image and text contributions without requiring per-dataset calibration.

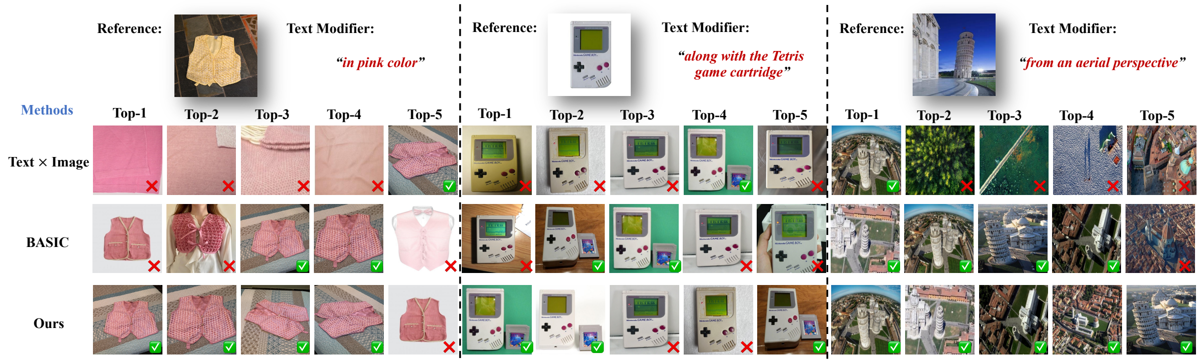


Figure 4: Qualitative retrieval examples on i-CIR. Each row shows a reference image with text modifier and the top-5 retrieved results for BASIC and MACE + CGCQE (ours). Ground-truth matches are highlighted with a green checkmark.

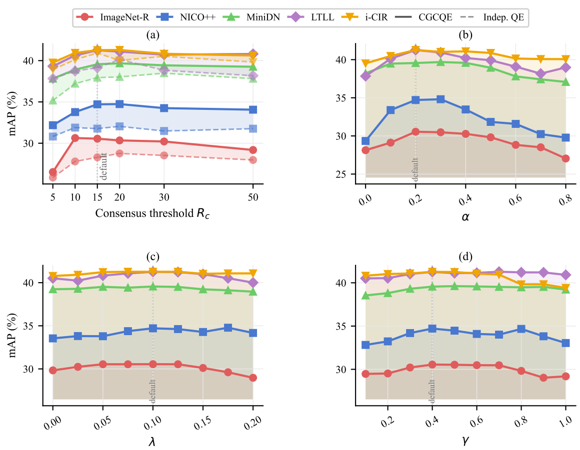


Figure 5: Sensitivity analysis of MACE + CGCQE. (a) Effect of consensus threshold R_c. Solid: CGCQE, dashed: per-granularity independent QE. (b)–(d) One-factor-at-a-time sweeps over α, λ, and γ.

Expansion weight γ. Panel (d) sweeps $\gamma \in [0.1, 1.0]$. Retrieval accuracy rises sharply from γ=0.1 and saturates around γ=0.4, showing that the consensus-expanded component meaningfully complements the original query without overwhelming it. At γ=1.0, where the query depends entirely on pseudo-relevant images, performance degrades noticeably. The broad plateau $\gamma \in [0.3, 0.6]$ provides ample operating room, and our default γ=0.4 lies well within this stable region.

4.6 Per-Category Performance on i-CIR

Figure 2 breaks down mAP on i-CIR by visual category and textual modification category, comparing MACE + CGCQE against BASIC [25] and MagicLens [39], the strongest trained competitor on i-CIR. MACE + CGCQE achieves the best performance in the majority of both visual and textual categories. The largest gains over BASIC are concentrated in visually fine-grained categories such as *fictional* characters and *product*, where the multi-granularity ensemble resolves subtle instance-level attributes that a single projection tends to conflate. For textual categories, our method consistently leads on *projection* and *appearance* modifications, which demand precise attribute-level discrimination. MACE + CGCQE surpasses MagicLens across all categories despite requiring no task-specific training, showing that spectral diversity transfers broadly across diverse query semantics.

4.7 Qualitative Case Study

Figure 4 shows representative retrieval examples from i-CIR, comparing MACE + CGCQE against BASIC. Each row presents a reference image and text modifier alongside the top-ranked results from each method. MACE + CGCQE more accurately satisfies both the visual identity constraint and the semantic modification specified by the query text. For fine-grained attribute changes such as viewpoint, context, or appearance, the multi-granularity ensemble captures complementary cues that a single projection misses, and the consensus-guided expansion reinforces query intent with high-confidence pseudo-relevant images rather than introducing outliers. Failure cases, where neither method retrieves the target, tend to involve extreme domain shifts or highly ambiguous textual modifiers, consistent with the per-category analysis.

5 Conclusion

We presented MACE with CGCQE for training-free composed image retrieval. MACE replaces fixed spectral truncation with a multi-granularity ensemble that fuses scores across nested contrastive PCA projections, capturing both category-level and instance-level similarities without modifying database embeddings. CGCQE strengthens pseudo-relevance feedback by selecting expansion candidates through cross-granularity consensus, preventing granularity-specific false positives from corrupting the expanded query. Together, these two components achieve state-of-the-art performance across all five benchmarks without any task-specific training. One limitation is that all hyperparameters are globally fixed. Future work could explore query-adaptive granularity selection for further gains.

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